

Unit 2, Lesson 2: Equivalent Conversions: Fractions ↔ Percentages – Part 1



Benchmark

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.



Learning Targets

- ☐ I can convert between fractions with denominators of 100 and percentages.
- ☐ I can convert between fractions and percentages using equivalent fractions.



2.2.1 Warm-Up: A Brief History of Numerical Systems

Humans went from using body parts, tally marks, and symbols to count objects, to what we now know as the base-ten system. Multiple civilizations around the world developed positional numbering systems, and both Mayan and Indian civilizations developed the concept of “zero.” The base-ten system was perfected by eighth century Indian mathematicians. They realized that they could combine the concept of “zero” with a positional system that used repeatable symbols, whose values are determined by their position in a sequence. A base of ten was selected because, quite simply, we have ten fingers. Centuries later, this Hindu-Arabic numeral system began to spread to Europe and replaced the Roman numeral system. This is what we use today.



1. What is the historical importance of the number zero?

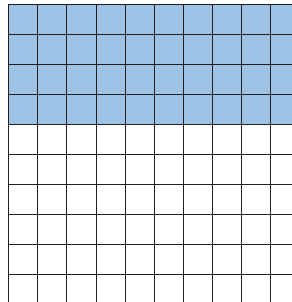
2. If zero was never created, how would that affect how we do math today?





2.2.2 Guided Instruction: Converting Between Fractions and Percentages

1. The model represents a numerical value. Use the model to answer the questions that follow.



- How many total squares are in the image?
- How many of the squares are shaded blue?
- What numerical value does the model represent, expressed as a fraction of $\frac{\text{blue shaded squares}}{\text{total squares}}$?

Recall that a **percentage** is a ratio expressed as a fraction of 100.

Percent (%) means “per 100,” “for every 100,” or “out of 100.”

For example, 40% means “40 out of 100,” so 40% is equivalent to the fraction $\frac{40}{100}$.

2. What decimal value has the same value as 40% and $\frac{40}{100}$?

Equivalent Values		
Percent Form	Fraction Form	Decimal Form
40%	$\frac{40}{100}$	

3. With your partner, complete the table by rewriting each value in the equivalent form indicated.

Fraction Form	Percent Form
$\frac{50}{100}$	50%
$\frac{71}{100}$	
$\frac{2}{100}$	
	150%
	42%

When converting fractions to percentages, generate an equivalent fraction with a denominator of 100 when possible.

4. Complete the table.

Original Fraction	Equivalent Fraction with a Denominator of 100	Percentage
$\frac{6}{10}$	$\frac{6}{10} \times \frac{10}{10} = \frac{60}{100}$	60%
$\frac{23}{50}$	$\frac{23}{50} \times \frac{2}{2} = \frac{\quad}{100}$	____%
$\frac{1}{5}$	$\frac{1}{5} \times \frac{\quad}{\quad} = \frac{\quad}{100}$	____%



2.2.3 Guided Practice: Converting Between Fractions and Percentages

1. Complete the table by rewriting each value in the equivalent form indicated. Simplify where applicable.

Fraction Form	Percent Form
$\frac{20}{50}$	40%
$\frac{13}{25}$	
$\frac{9}{10}$	
	30%
	18%
$\frac{5}{4}$	
	200%



2.2.4 Your Turn!

1. Rewrite each fraction as an equivalent percentage.

a. $\frac{17}{100}$

b. $\frac{42}{50}$

c. $\frac{36}{25}$

d. $\frac{96}{100}$

e. $1\frac{9}{100}$

f. $\frac{24}{20}$

g. $\frac{1}{50}$

h. $\frac{4}{5}$

2. Convert each to a fraction. Simplify where applicable.

a. 90%

b. 167%

c. 86%

d. 33%

e. 226%

f. 29%

g. 50%

h. 18%



2.2.5 Wrap-Up: Ticket Out the Door

- Use the numbers below to create three more fractions equivalent to any three percentages in the table.

15	4	21	30	25	40
32	1	8	2	6	10
7	50	72	5	80	3

10%	20%	30%	40%	50%	60%	70%	80%	90%
								$\frac{72}{80}$

Unit 2, Lesson 3: Equivalent Conversions: Fractions ↔ Percentages – Part 2



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Learning Targets

- ☐ I can convert between fractions and percentages.
- ☐ I can rewrite positive rational numbers in different but equivalent forms.